

SECURE ENTERPRISE NETWORK LAB

Cisco Packet Tracer Network Design & Implementation

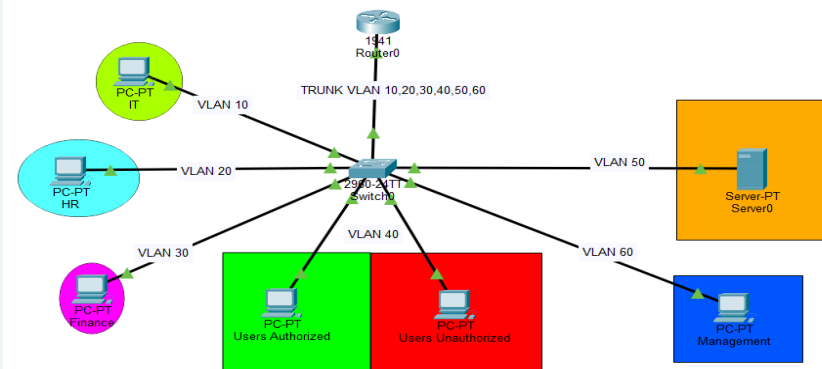
PROJECT OVERVIEW

A secure single-site enterprise network designed and implemented in Cisco Packet Tracer. The project demonstrates VLAN segmentation, inter-VLAN routing, DHCP, DNS, FTP, TFTP, ACLs, port security, and SSH for secure network management and controlled access.

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CISCO NETWORK PROJECT

Implementation, configuration and verification



01 INITIAL TOPOLOGY

CONFIGURATION / ACTION

The enterprise network topology showing the router, switch, endpoints, and server.

Packet Tracer topology

Switch0

02 VLAN CONFIGURATION

CONFIGURATION / ACTION

Created VLANs and assigned departments to the correct access ports.

```
vlan 10,20,30,40,50,60
name IT, HR, Finance, Users, Servers, Management
```

Implementation, configuration and verification



CONFIGURATION / ACTION

Configured Gi0/1 as an 802.1Q trunk.

```
interface Gi0/1
switchport mode trunk
```



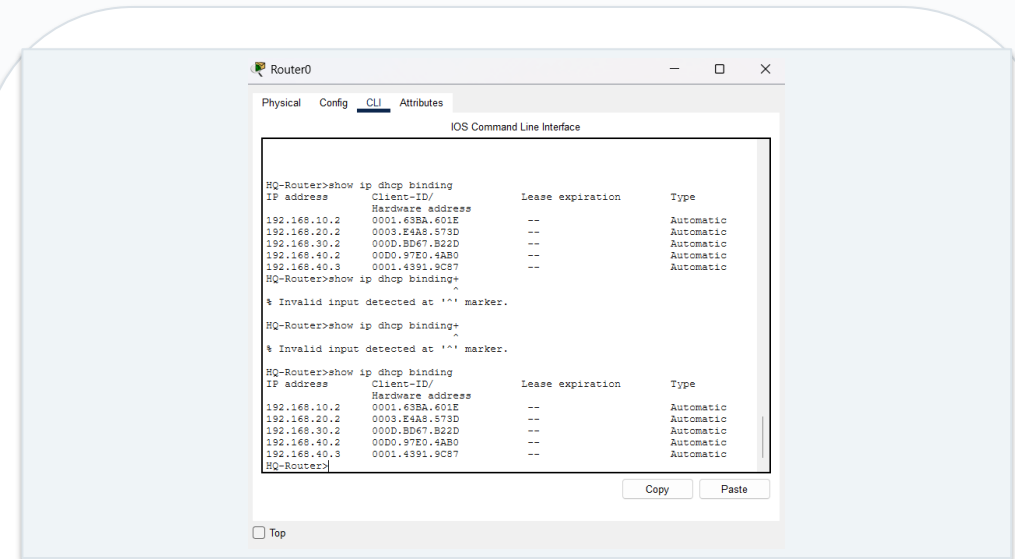
CONFIGURATION / ACTION

Configured subinterfaces for inter-VLAN routing.

```
interface Gi0/0.X
encapsulation dot1Q X
ip address 192.168.X.1 /24
```

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Implementation, configuration and verification

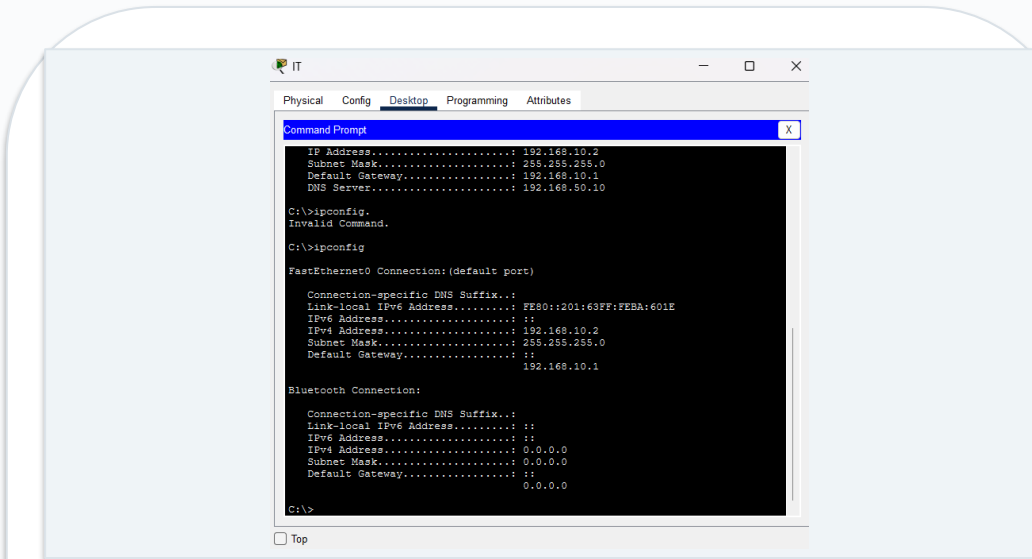


05 DHCP VERIFICATION

CONFIGURATION / ACTION

Configured DHCP for automatic IP assignment.

```
ip dhcp pool VLAN10-IT
network 192.168.10.0 /24
default-router 192.168.10.1
dns-server 192.168.50.10
```



06 IP ADDRESSING

CONFIGURATION / ACTION

IT PC received its IP configuration automatically.

DHCP

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Implementation, configuration and verification

```

Switch0
HQ-Switch#show ip interface brief
Interface      IP-Address      OK? Method Status
Protocol
FastEthernet0/1 unassigned      YES manual up
FastEthernet0/2 unassigned      YES manual up
FastEthernet0/3 unassigned      YES manual up
FastEthernet0/4 unassigned      YES manual up
FastEthernet0/5 unassigned      YES manual up
FastEthernet0/6 unassigned      YES manual up
FastEthernet0/7 unassigned      YES manual up
FastEthernet0/8 unassigned      YES manual down
FastEthernet0/9 unassigned      YES manual down
FastEthernet0/10 unassigned      YES manual down
FastEthernet0/11 unassigned      YES manual down
FastEthernet0/12 unassigned      YES manual down
FastEthernet0/13 unassigned      YES manual down
FastEthernet0/14 unassigned      YES manual down
FastEthernet0/15 unassigned      YES manual down
FastEthernet0/16 unassigned      YES manual down
FastEthernet0/17 unassigned      YES manual down
FastEthernet0/18 unassigned      YES manual down
FastEthernet0/19 unassigned      YES manual down
FastEthernet0/20 unassigned      YES manual down
FastEthernet0/21 unassigned      YES manual down
FastEthernet0/22 unassigned      YES manual down
FastEthernet0/23 unassigned      YES manual down
FastEthernet0/24 unassigned      YES manual down
GigabitEthernet0/1 unassigned      YES manual up
GigabitEthernet0/2 unassigned      YES manual down
Vlan1 unassigned      YES manual administratively down
Vlan60 192.168.60.3 YES manual up
  
```

07 MANAGEMENT VLAN

CONFIGURATION / ACTION

Configured VLAN 60 for switch management.

```

interface vlan 60
interface VLAN 60
ip address 192.168.60.3 /24
no shutdown
ip default-gateway 192.168.60.1
  
```

```

Switch0
HQ-Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
          (Count)          (Count)          (Count)
-----
Fa0/1      1            1            0            Shutdown
Fa0/2      1            1            0            Shutdown
Fa0/3      1            1            0            Shutdown
Fa0/4      1            1            0            Shutdown
Fa0/5      1            1            0            Shutdown
Fa0/6      1            1            0            Shutdown
Fa0/7      1            1            0            Shutdown

HQ-Switch#show port-security address
Secure Mac Address Table
-----
Vlan  Mac Address      Type      Ports      Remaining
Age                                     (mins)
-----
10     0001.69BA.601E      SecureSticky Fa0/1      -
20     0003.FA9A.579D      SecureSticky Fa0/2      -
30     000D.BD67.B22D      SecureSticky Fa0/3      -
40     000D.97E0.4AB0      SecureSticky Fa0/4      -
40     0001.4391.9C87      SecureSticky Fa0/5      -
60     0060.3E42.D8E9      SecureSticky Fa0/6      -
50     00E0.F91E.46B3      SecureSticky Fa0/7      -

Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024
HQ-Switch#
  
```

08 PORT SECURITY & STICKY MAC

CONFIGURATION / ACTION

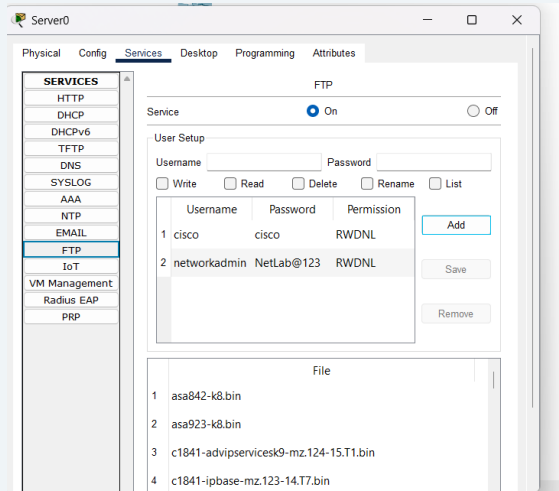
Restricted each access port to one authorized MAC address.

```

switchport port-security
maximum 1
sticky MAC
violation shutdown
  
```

CISCO NETWORK PROJECT

Implementation, configuration and verification

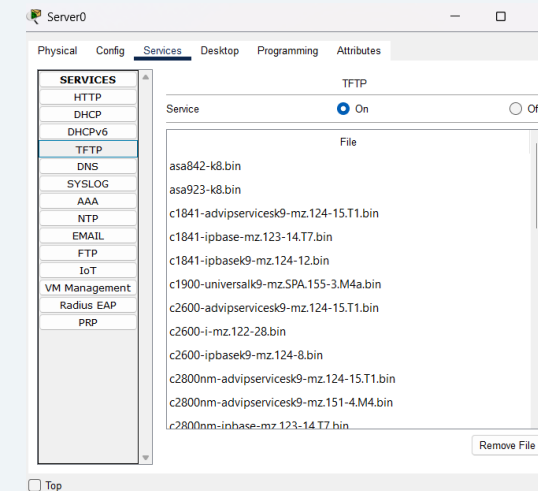


09 FTP SERVICE

CONFIGURATION / ACTION

Enabled FTP for file transfer.

```
FTP → On
Username: networkadmin
Password: NetLab@123
```



10 TFTP SERVICE

CONFIGURATION / ACTION

Enabled TFTP for configuration backups.

```
TFTP → On
```

CISCO NETWORK PROJECT

Implementation, configuration and verification

```

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping server

Pinging 192.168.50.10 with 32 bytes of data:
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127
Reply from 192.168.50.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
  
```

11 DNS / SERVER CONNECTIVITY

CONFIGURATION / ACTION

Configured DNS and verified server connectivity.

DNS → On
A Record: server → 192.168.50.10

```

C:\>ping 192.168.60.1

Pinging 192.168.60.1 with 32 bytes of data:
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255
Reply from 192.168.60.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.60.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.60.3

Pinging 192.168.60.3 with 32 bytes of data:
Request timed out.
Reply from 192.168.60.3: bytes=32 time<1ms TTL=255
Reply from 192.168.60.3: bytes=32 time<1ms TTL=255
Reply from 192.168.60.3: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.60.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
  
```

12 MANAGEMENT CONNECTIVITY

CONFIGURATION / ACTION

Verified connectivity to the switch management interface.

ping 192.168.60.3

CISCO NETWORK PROJECT

Implementation, configuration and verification

```

C:\>ping 192.168.40.1
Pinging 192.168.40.1 with 32 bytes of data:
Reply from 192.168.40.1: bytes=32 time=1ms TTL=255
Reply from 192.168.40.1: bytes=32 time=1ms TTL=255
Reply from 192.168.40.1: bytes=32 time=1ms TTL=255
Reply from 192.168.40.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.2
Pinging 192.168.10.2 with 32 bytes of data:
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127
Reply from 192.168.10.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
  
```

13 UNAUTHORIZED ACCESS TEST

CONFIGURATION / ACTION

Blocked unauthorized access to the server.

```
deny ip host 192.168.40.3 host 192.168.50.10
permit ip any any
ip access-group USERS-TO-SERVER in
```

```

C:\>ftp 192.168.50.10
Trying to connect...192.168.50.10
Connected to 192.168.50.10
220- Welcome to FT Ftp server
Username:networkadmin
331- Username ok, need password
Password:*****
230- Logged in
ftp>
!Error opening ftp://192.168.50.10/ (Timed out)
.

(Disconnecting from ftp server)

C:\>ssh -l admin 192.168.60.1
Password:
HQ-Router#
  
```

14 SSH ROUTER

CONFIGURATION / ACTION

Enabled secure SSH management for the router.

```
ip domain-name netlab.local
crypto key generate rsa
username admin privilege 15 secret NetLab@123
line vty 0 4
login local
transport input ssh
ip ssh version 2
```